Overview of Python

Python is a high-level, interpreted, object-oriented programming language known for its simple syntax and readability. It supports multiple programming paradigms such as procedural, object-oriented, and functional programming.

Features of Python:

- Easy to learn and use
- Interpreted language
- Platform independent
- Large standard library
- Supports dynamic typing

Applications of Python:

- Web Development
- Data Science & Machine Learning
- Artificial Intelligence
- Automation & Scripting
- Game Development

Basic Syntax of Python

1. Python uses indentation instead of braces {}.
2. Code blocks are defined using consistent spacing.
3. Python is case-sensitive.
4. Comments are written using #.

Printing Syntax

```
print("Hello, World!")
```

```
Hello, World!
```

Python Variables

```
x = 10                          # Integer
y = 3.14                        # Float
name = "Jashith"                # String
is_active = True                # Boolean
complex_num = 10+3j             # Complex
print(f"x: {x}, type: {type(x)}")
print(f"y: {y}, type: {type(y)}")
print(f"name: {name}, type: {type(name)}")
print(f"is_active: {is_active}, type: {type(is_active)}")
print(f"complex_num: {complex_num}, type: {type(complex_num)}")
```

```
x: 10, type: <class 'int'>
y: 3.14, type: <class 'float'>
name: Jashith, type: <class 'str'>
is_active: True, type: <class 'bool'>
complex_num: (10+3j), type: <class 'complex'>
```

Python Local Variables

```
def sum(x,y):
   sum = x + y
   return sum
print(sum(5, 10))
```

```
15
```

Python Global Variables

```
x = 5
y = 10

def sum():
   sum = x + y
   return sum
print(sum())
```

```
15
```

Operators

1. Arithmetic Operators

```
a = 10
b = 3
print("\nArithmetic Operators:")
print(f"Addition (a + b): {a + b}")
print(f"Subtraction (a - b): {a - b}")
print(f"Multiplication (a * b): {a * b}")
print(f"Division (a / b): {a / b}")
print(f"Floor Division (a // b): {a // b}")
print(f"Modulus (a % b): {a % b}")
print(f"Exponentiation (a ** b): {a ** b}")
```

```
Arithmetic Operators:
Addition (a + b): 13
Subtraction (a - b): 7
Multiplication (a * b): 30
Division (a / b): 3.3333333333333335
Floor Division (a // b): 3
Modulus (a % b): 1
Exponentiation (a ** b): 1000
```

2. Comparison Operators (Return True/False)

```
print("\nComparison Operators:")
print(f"Equal (a == b): {a == b}")
print(f"Not Equal (a != b): {a != b}")
print(f"Greater than (a > b): {a > b}")
print(f"Less than (a < b): {a < b}")
print(f"Greater than or equal to (a >= b): {a >= b}")
print(f"Less than or equal to (a <= b): {a <= b}")
```

```
Comparison Operators:
Equal (a == b): False
Not Equal (a != b): True
Greater than (a > b): True
Less than (a < b): False
Greater than or equal to (a >= b): True
Less than or equal to (a <= b): False
```

3. Logical Operators

```
print("\nLogical Operators:")
var = 5

print(var > 3 and var < 10)
print(var > 3 or var < 4)
print(not (var > 3 and var < 10))
```

```
Logical Operators:
True
True
False
```

4. Assignment Operators

```
c = 5
print("\nAssignment Operators:")
c += 2
print(f"c += 2 -> {c}")
c -= 2
print(f"c -= 2 -> {c}")
c *= 2
print(f"c *= 2 -> {c}")
c /= 2
print(f"c /= 2 -> {c}")
c //= 2
print(f"c //= 2 -> {c}")
c %= 2
print(f"c %= 2 -> {c}")
c **= 2
print(f"c **= 2 -> {c}")
```

```
Assignment Operators:
c += 2 -> 7
c -= 2 -> 5
c *= 2 -> 10
c /= 2 -> 5.0
c //= 2 -> 2.0
c %= 2 -> 0.0
c **= 2 -> 0.0
```

5. Bitwise Operators

```
print("\nBitwise Operators:")

print("\nBitwise AND Operator:")
print(f"a & b: {a & b}")

print("\nBitwise OR Operator:")
print(f"a | b: {a | b}")

print("\nBitwise XOR Operator:")
print(f"a ^ b: {a ^ b}")

print("\nBitwise NOT Operator:")
```

```
print(f"~a: {~a}")

print("\nBitwise Left Shift Operator:")
print(f"a << 2: {a << 2}")

print("\nBitwise Right Shift Operator:")
print(f"a >> 2: {a >> 2}")
```

```
Bitwise Operators:

Bitwise AND Operator:
a & b: 2

Bitwise OR Operator:
a | b: 11

Bitwise XOR Operator:
a ^ b: 9

Bitwise NOT Operator:
~a: -11

Bitwise Left Shift Operator:
a << 2: 40

Bitwise Right Shift Operator:
a >> 2: 2
```